Adaptive AI Technical Interviewer

Final report

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Code: `https://github.com/Srinjoy003/LMA-RAG`

Abstract

We built an interviewer for junior technical screening in Python, SQL and data structures. It keeps an estimate of the candidate's level in each skill, picks each question to reduce its uncertainty, stops when it is confident, and works out *why* a wrong answer is wrong, including which misconception the candidate holds. It recommends fit, not fit or human review and never makes the hiring decision. In the final experiment five systems interviewed the same 22 simulated candidates (110 interviews), with Gemini grading and a different model family playing the candidate. **Belief modelling:** the full system confirms fewer planted misconceptions than a version without the belief model (34% against 56%), but its findings are far more trustworthy: 58% of what it confirms is a planted misconception, against 20% (difference +38.3 points, 95% interval +18.7 to +62.3), and it never named a misconception on a correct answer. **Skill levels:** adaptive question choice at the fixed interview's length judged 62% of levels correctly against 52% (+10.6 points, interval −4.5 to +25.8). The full system did not beat the fixed interview on levels (53%) and asked more (20.1 exchanges against 12.0). We trace this to a trade-off built into the design: to confirm a belief it re-asks about it, and those deliberately wrong answers pull the level estimate down.

1 Introduction

A screening interview has to judge several skills with a handful of questions. Fixed question lists waste questions on skills that are already clear and leave others under-tested. LLM interviewers make screening cheap, but they decide what to ask and when to stop from a prompt alone, with no record of how sure they are. Adaptive testing solves this for multiple-choice tests by asking the most informative question and stopping once the estimate is precise [8], and TestAgent [7] brings the idea to conversational assessment. We apply it to open answers across several skills, and add a model of what the candidate *believes*: which known misconception, if any, explains a wrong answer. Our two research questions:

RQ1 Can uncertainty-driven questioning with early stopping match a fixed 12-question interview's accuracy with fewer exchanges?

RQ2 Does modelling what the candidate believes find their misconceptions better than a system without that model?

Main findings.

- **The belief model makes misconception findings trustworthy (RQ2).** With it, 58% of confirmed misconceptions are planted ones; without it, 20%. The price is lower recall (34% against 56%).
- **Adaptive question choice helps at equal length (RQ1).** At exactly 12 questions it judged 62% of levels correctly against 52% for the fixed list; the interval includes a small loss, so this is a likely gain, not a proven one.
- **The full system is not more accurate on levels and is longer (RQ1).** It judged 53% of levels correctly with 20.1 exchanges. Section 6.3 shows why: the belief model's probing questions cost level accuracy.
- **Safe decisions.** No adaptive system passed an unfit candidate or rejected a fit one; the fixed interview passed one unfit candidate.

2 What changed since the mid-term report

Part	Mid-term	Final
Question bank	135 drafted questions, a quarter reviewed	all 135 reviewed by the team; 43 corrected
Belief model	A2: a misconception confirmed by two replies, which could be one question and its own follow-up	A2E: two *different* questions, each backed by a quote from the candidate's answer (Section 3.1)
Systems run	B1 and A2	B1, A2E, A4, ADAPT12, A5E on the same candidates
Equal-length control	stub candidates only	live (ADAPT12)
Run records	results folders	a manifest per run that fingerprints code, bank, prompts and setup

The mid-term found that A2's confirmed misconceptions were mostly not planted ones (18% precision), largely because a main answer and its own follow-up counted as two confirmations. The evidence rules in A2E are the response to that finding.

3 The system

Figure 1 shows the loop. The LLM is used only where language is needed. Everything that controls the interview (which question, when to stop, the verdict rule) is plain code, so the candidate cannot change the procedure by asking.

Scorecard. Each skill has an ability value θ on a logit scale. Questions come in three levels with difficulty $b = -1, 0, +1$. An answer earns 0, 1 or 2 marks, and we model the chance of each mark with the Partial Credit Model [5]:

$$P(X = x \mid \theta, b) \;\propto\; \exp\Big(\sum_{k \le x} (\theta - \delta_k)\Big), \qquad \delta_1 = b - 0.5,\ \delta_2 = b + 0.5.$$

We start from a normal prior (mean 0, standard deviation 1) and update it on a grid after every graded question. The estimate is the posterior mean (EAP, [2]) and the uncertainty is the posterior standard deviation (SE). Each answer is graded three times; if the marks disagree, it counts as weaker evidence (2 of 3 agreeing gives weight 0.67). The estimate becomes a level 0–3 with cut points at $\theta = -1.25$, -0.25 and 0.75, set on stub runs in the mid-term work so that the fixed baseline is not biased upwards.

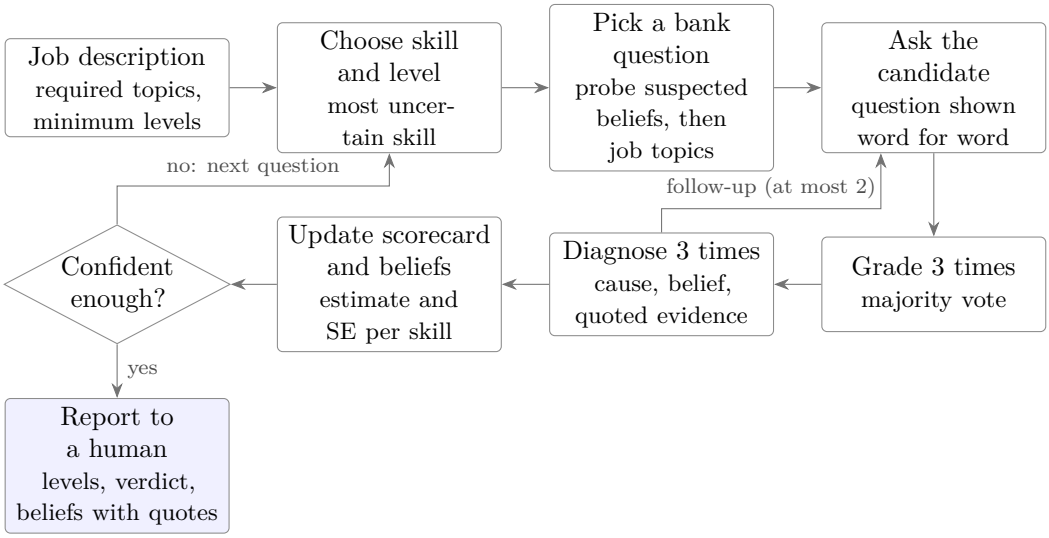


Figure 1: One interview of the full system (A2E). The LLM grades, diagnoses and writes follow-up questions; code makes every decision (which question, when to stop, the verdict).

Choosing the next question. The next skill is the one with the highest SE, and the level is the one whose difficulty is closest to that skill's estimate. Among the unasked bank questions of that skill and level, the system prefers, in order: a question listing a misconception it already suspects; a question on a topic the candidate just got wrong; a question on a topic the job needs that has not been tested yet. Remaining ties are broken by a seeded random choice.

Stopping. The interview stops when every skill has at least two answers and an SE below 0.6, or after 15 main questions. We chose 0.6 for accuracy rather than length, as the mid-term report explains.

Grading and follow-ups. The grader marks the main answer three times. A follow-up is asked when the diagnosis is not a plain knowledge gap, at most two per question. The grader then marks the *whole exchange* three times, and that final grade updates the scorecard. A reply that shows a listed misconception scores 0 even if other parts are right.

Verdict. For each skill the system computes the posterior probability that θ is at or above the job's minimum level. Above 0.8 the skill meets the minimum, below 0.2 it does not, otherwise it is uncertain. The verdict is *not fit* if any skill is clearly below, *fit* if all clearly meet, and *needs human review* otherwise.

3.1 Modelling what the candidate believes

Beside the skill scorecard, the full system keeps a model of what the candidate believes, a simple theory of mind. A wrong answer is not only a low mark: the diagnoser decides *why* it is wrong (a gap in knowledge, a misread, a slip, or a specific wrong belief from the question's list), and the system tracks each belief across the interview. In A2E this works as follows.

1. **Three votes with evidence.** The diagnoser runs three times. A vote for a misconception must quote the words in the candidate's answer that show it and say why. Code checks that the quote really is in the answer; a vote without a valid quote counts as a knowledge gap. A misconception needs a strict majority of the three votes.
2. **Suspected, then confirmed.** One supported reply makes a belief *suspected*; the system then prefers a later question that lists it. The belief is *confirmed* only when it is supported on two different main questions. A question and its own follow-up count once.

3. **Contradictions.** Correct answers are diagnosed too. If the candidate later states the opposite of a suspected belief, with a quote, the finding becomes *disputed* and is not reported as confirmed.

Nothing here is learned; it is prompts plus counting rules in code (`src/part3/diagnoser.py`, `src/engine/followups.py`, `src/shared/evidence.py`). The ablation **A4** removes it: one diagnosis on the main answer, no quotes, generic follow-ups, and confirmation from a single reply. Comparing A2E with A4 answers RQ2.

A correction before the final run. The quote check first required an exact character match. In the first final-run attempt we found that the candidate model writes narrow no-break spaces and backticks around code, which Gemini drops when it quotes, so correct diagnoses with correct quotes were being discarded. In a replay of that attempt's diagnoses, 32 of the 150 misconception votes (21%) were rejected for formatting alone. We changed the check to ignore spaces, backticks, quotation marks and letter case; the quote must still be the candidate's own words, and the stored evidence is the exact span of the answer. The attempt was discarded and the final run restarted under a new version with this fix (Section 5).

3.2 Tool calling

The engine can also be run by an LLM agent instead of the code loop. The agent calls six tools (`search_question_bank`, `ask_candidate`, `grade_answer`, `ask_follow_up`, `update_scorecard`, `end_interview`), and each tool enforces the same rules as the code driver. A demonstration (`scripts/agent_demo.py -probe-rules`) runs a full interview and deliberately breaks the rules: ending before the stop rule is met, asking a question the rules did not choose, updating the scorecard before grading, moving on while a question is still open, and calling a tool that does not exist. All five are refused with a reason, the agent's written summary agrees with the computed decision, and the agent reaches the same questions, levels and verdict as the code driver. The experiments use the code driver, which gives the same result with about 55 fewer model calls per interview.

4 Data

Question bank. 135 questions: 3 skills × 3 levels × 15, each with a model answer, key points, a 0/1/2 rubric, topic tags and a list of common misconceptions. There are *explain* and *read-code* questions. Questions were drafted with an LLM; every question was then reviewed by a teammate for answer correctness, rubric consistency, relevant misconceptions and level. The review flagged 66 questions and we corrected 43 (wrong claims in model answers, unclear wording, and misconceptions listed on questions that do not test them). The other notes asked for rubric changes our grading rule already covers. Level notes were resolved by aligning the written level definitions with the topic levels the bank was built from, without moving questions.

Misconceptions. 95 misconceptions. Where a published catalogue exists we adapted from it: Progmiscon for Python [3], a think-aloud study of SQL novices [6], concept-inventory work on data structures [4] and the McMining benchmark [1]. The rest were written for this project.

Jobs. Three job descriptions (junior data analyst, backend engineer, data engineer), each with a minimum level per skill and 13 to 17 required topics.

Simulated candidates. Each candidate has a hidden answer key: a level (0–3) for every topic, one or two planted misconceptions, and small chances of misreading or slipping. A candidate's true level for a skill is the floor of their mean level over the topics the job requires; planted misconceptions do not change it. Candidates come in blocks of 12 that cover every combination

of job, even or uneven profile, and fit or not fit once. For every question, code decides what the candidate should do (answer correctly, partly, wrongly, with a planted misconception, misread, or slip) and a second LLM writes the answer following that plan.

5 Experimental setup

System	What it is	Compared with
B1	Fixed: 12 questions (levels 1, 2, 2, 3 per skill), no follow-ups, no diagnosis	–
ADAPT12	Adaptive question choice, exactly 12 questions, no follow-ups, no diagnosis	B1 (RQ1, equal length)
A2E	Full system: adaptive choice, early stopping, belief model with evidence	B1 (RQ1), A4 (RQ2)
A4	A2E without the belief model (single diagnosis, no quotes, one-reply confirmation)	A2E (RQ2)
A5E	A2E with job-blind question choice	A2E (job-aware retrieval)

Table 1: The five systems. Every system interviewed every candidate.

Models. Grading, diagnosis and topic extraction use `gemini-3.5-flash-lite`; the simulated candidate is `gpt-oss-120b` on Groq, a different model family. Before the run, one interview each of B1 and A2E on a separate validation candidate checked that every grade and diagnosis got three valid votes.

Runs. Candidates were run one at a time through all five systems, in the order of the balanced blocks, so that a run stopped early still holds complete, paired candidates. The run stopped after 22 complete candidates (110 interviews): the free Groq keys run out each day, and the prepaid Gemini credit (about Rs. 300) ran out on the day of the run. The first block of 12 is complete and balanced; the second is missing two candidates. Every comparison below uses the same 22 candidates in every system. The run's manifest fingerprints the source code, bank, prompts, models and candidate files, and results from a different setup cannot be mixed in.

Metrics. *Skill accuracy* is the share of (candidate, skill) pairs where the reported level equals the true level; we also report the mean absolute level error. *Exchanges* are main questions plus follow-ups. *Verdicts* are right, wrong, or sent to a human. *Misconception recall* is the share of planted misconceptions that were confirmed, and *precision* the share of confirmed misconceptions that were planted. The *false-alarm rate* is the share of replies planned as correct on which a misconception was named.

Uncertainty. With 22 candidates the numbers move a lot from sample to sample, so we give 95% bootstrap intervals: candidates are redrawn with replacement 2,000 times, with the same draw for every system because they interviewed the same people. For a comparison the interval is for the difference between the two systems.

6 Results

6.1 RQ1: skill levels and length

Adaptive choice at equal length. ADAPT12 and B1 both ask exactly 12 questions; only the choice of questions differs. ADAPT12 judged 62% of levels correctly against 52%, a difference of +10.6 points [−4.5, +25.8]. The interval includes a small loss, so we read this as a likely gain from

	B1	ADAPT12	A2E	A5E	A4
Skill levels right	52%	62%	53%	62%	70%
Mean level error	0.52	0.39	0.48	0.38	0.30
Exchanges per interview	12.0	12.0	20.1	21.3	18.9
Verdicts right / to a human / wrong	9 / 12 / 1	10 / 12 / 0	9 / 13 / 0	11 / 11 / 0	14 / 8 / 0
Misconception recall	–	–	34%	38%	56%
Misconception precision	–	–	58%	57%	20%
False alarms on correct replies	–	–	0.0%	0.7%	0.0%
Required job topics tested	90%	76%	74%	62%	85%
Interviewer LLM calls	37	37	113	120	65

Table 2: All five systems on the same 22 candidates (32 planted misconceptions).

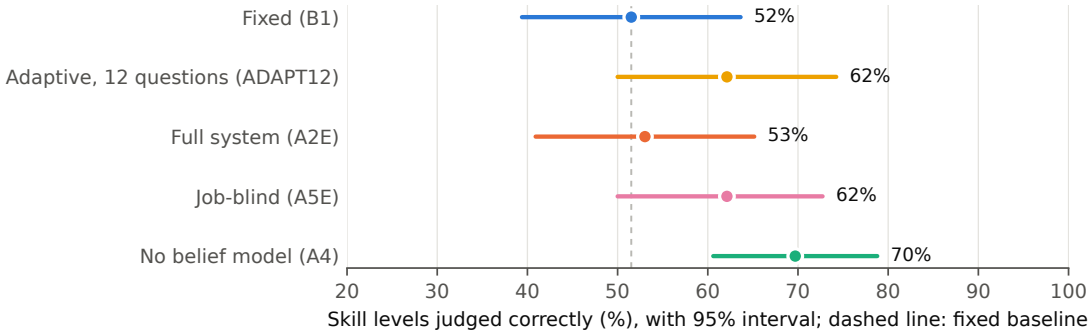


Figure 2: Skill levels judged correctly, with 95% intervals over candidates.

adaptivity rather than a proven one. It repeats the direction of the mid-term stub controls, now with the live models.

The full system against the fixed interview. A2E judged 53% of levels correctly, a difference of +1.5 points from B1 [−16.7, +19.7]: no measurable gain. It asked 20.1 exchanges per interview against 12.0 (difference +8.1 [+7.0, +9.4]), mostly follow-ups (Figure 3). RQ1's target of 25% fewer exchanges is not met. Its errors are not one sided: 17 levels too high and 14 too low, where B1 errs mostly upwards (24 and 8).

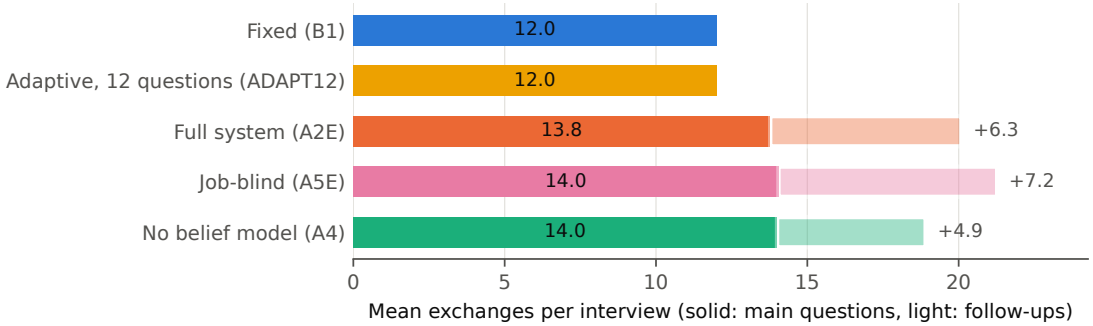


Figure 3: Interview length: main questions plus follow-ups.

Verdicts. No adaptive system passed an unfit candidate or rejected a fit one; B1 passed one unfit candidate (Figure 4). All systems send many candidates to a human (between a third and 60%), because the 0.8 rule demands a clear margin above the job minimum. A4 decided the most cases (14 of 22), in line with its better level estimates.

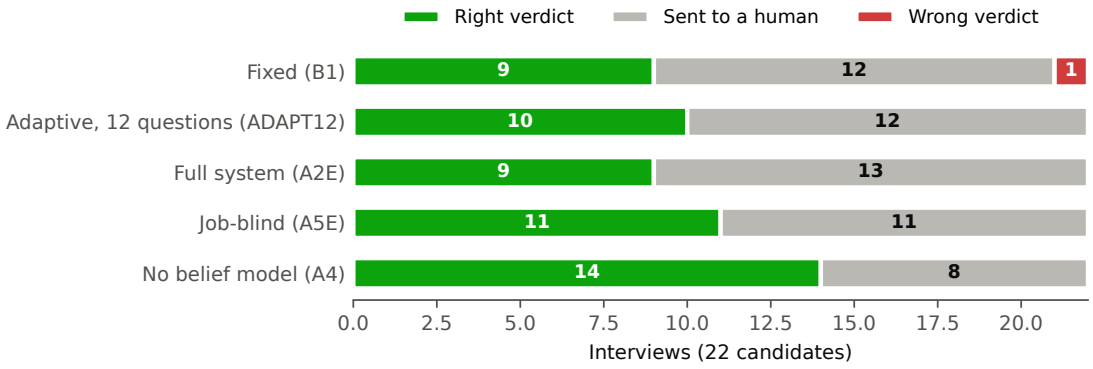


Figure 4: Fit / not-fit verdicts per system.

6.2 RQ2: does modelling beliefs help?

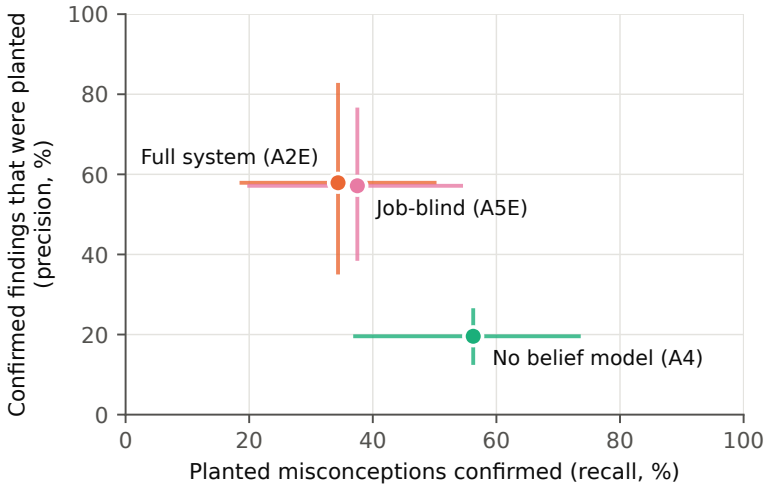


Figure 5: Recall against precision of confirmed misconceptions, with 95% intervals. Up is more trustworthy, right finds more.

Figure 5 shows the trade-off. Without the belief model, A4 confirmed 92 misconceptions in 22 interviews and found 18 of the 32 planted ones (recall 56%), but only 20% of its findings were planted. With the belief model, A2E confirmed 19 and found 11 (recall 34%), and 58% of its findings were planted:

A2E − A4	Difference	95% interval
Precision	+38.3 points	+18.7 to +62.3
Recall	−21.9 points	−35.5 to −9.4
False alarms on correct replies	+0.0 points	+0.0 to +0.0

Both differences are clear. For a screening tool, precision matters more: telling a candidate they hold a misconception they do not is worse than missing one, and A4 is wrong about four times in five. A2E also kept 59 beliefs as suspected (seen on one question only) and marked 2 as disputed after the candidate contradicted them; neither counts as a finding. Neither system named a misconception on a correctly answered reply. The recall target of 60% from the proposal is not met. Precision here is against planted misconceptions only: a simulated candidate told to answer wrongly sometimes states a common misconception that was not planted, so some "false" findings may be right about the reply.

6.3 Why the full system is less accurate on levels than A4

A4 judged 70% of levels correctly and A2E 53%, a difference of −16.7 points [−28.8, −4.5]. Both use the same scorecard, question-choice rules and stopping rule, so the difference comes from the belief model. To confirm a belief on a second question, A2E prefers questions that list a suspected misconception. A candidate who holds it answers those wrongly by design. A2E asked 1.5 such questions per interview, A4 0.8 and B1 0.5. Each scores 0 and pulls the estimate down, but the true level ignores planted misconceptions, so the result shows up as under-estimates: 14 of A2E's wrong levels are too low, against 4 for A4. This is a real cost of probing beliefs, partly made larger by our definition of the true level. A system that wants both could keep the probing questions out of the level estimate, or stop probing once a belief is confirmed.

6.4 Premature stops

An interview stops early when every skill's SE is below 0.6. A2E stopped this way in 20 of 22 interviews, and after those stops 29 of 60 skill levels were wrong (A4: 15 of 45).

- **By skill:** Python 13, data structures 10, SQL 6 wrong levels out of 20 each. SQL is judged best.
- **By size:** all but 1 of the wrong levels after an early stop are one level off; the rule stops as soon as the estimate is precise enough, even when it sits next to a cut point.
- **At the bottom of the scale:** 2 of the 5 skills a candidate did not know at all (true level 0) were reported as level 1.

SE below 0.6 means the estimate is precise, not that it is far from a cut point. Asking one more question when the estimate is close to a cut point, or requiring most of the posterior to lie in one level, would address this; the second rule was implemented (A2EB) but not run live.

6.5 Job-aware question choice

A5E differs from A2E only in ignoring the job description when choosing questions. It tested fewer of the job's required topics (62% against 74%, a difference of −11.4 points [−16.7, −5.8]), which is what job-aware retrieval is for. Its level accuracy and misconception recall were not measurably different (+9.1 points [−3.0, +21.2] on levels). So job-aware choice makes the interview cover the job, without a measurable effect on accuracy at this sample size.

6.6 Cost

A2E made 113 interviewer-side LLM calls per interview against 37 for B1: three grading and three diagnosis votes per answer, including correct ones. A4 needed 65. The experiment, including the discarded first attempt, used about Rs. 300 of Gemini credit; the candidate model ran on Groq's free tier.

7 Discussion

What we learned. Adaptive question choice does what adaptive testing promises: at the same length it places candidates better than a fixed list, though our sample cannot prove it. The belief model does what it was redesigned to do: its findings are trustworthy where the mid-term version's were not. The two do not combine for free. Confirming a belief needs evidence from a second question, and that evidence comes from deliberately probing a weakness, which costs level accuracy and length. A2E is therefore not the best system on every measure: A4 judges levels best, ADAPT12 is the best short interview, and A2E gives the most trustworthy explanation of *why* a candidate is wrong.

	Target	Result	Status
RQ1	Skill accuracy within 5 points of B1	+1.5 points [−16.7, +19.7]	not settled
RQ1	At least 25% fewer exchanges than B1	20.1 against 12.0	not met
RQ1	Adaptive choice beats the fixed list at equal length	+10.6 points [−4.5, +25.8]	not settled
RQ2	More trustworthy findings than without the belief model	+38.3 points precision	met
RQ2	Misconception recall of at least 60%	34%	not met
RQ2	False alarms on correct answers below 15%	0.0%	met

Table 3: The targets against the final results.

Mid-term against final. In the mid-term, A2 beat B1 on levels by 17 points. With A2E on the reviewed bank and new candidates it does not. Two things changed: the stricter belief model (which probes more, as shown above) and the corrected question bank; the candidates are the same people (the first 22 of the 24 used then). The mid-term gain also included more questions, as that report noted, and our final equal-length comparison is the cleaner test of adaptivity.

The evidence check matters. A rule that checks the model's evidence is only as good as the check. An exact-match rule silently threw away a fifth of correct diagnoses because of invisible formatting differences between two models. Checking evidence against real outputs, not only test cases, caught it.

8 Limitations

- **Simulated candidates.** Real candidates are less consistent. The simulated candidate sometimes ignored its plan (graded answers matched the plan for 87% of A2E's questions), and a follow-up occasionally gave the answer away, after which the grader gave full marks. Results describe the interviewer against a controlled population.
- **Sample size.** 22 candidates give wide intervals; several comparisons are not settled. The run stopped at 22 of 60 candidates when the API budget ran out.
- **Truth ignores misconceptions.** A candidate's true level does not count planted misconceptions, which favours systems that do not probe them (Section 6.3).
- **One grader model, no human check yet.** Every grade comes from one Gemini model. The planned hand checks were not done: two teammates independently marking a sample of answers against the rubric, and labelling 30 simulated answers against their plans, reporting Cohen's κ between them with $\kappa \geq 0.6$ as acceptable. If κ is lower, the labelling guide is ambiguous: we would rewrite it, settle the disagreements together and label a fresh sample. If the two agree with each other but not with the model, the fault is the model's: we would report the grader's error rate beside every accuracy figure, and score simulated answers against what was written rather than the plan.
- **Settings tuned before A2E.** The stopping threshold and cut points were tuned on stub runs before the belief model was changed.
- **Safety.** Only one manipulation attempt per system was in this sample. It did not raise the grade and was flagged, but one case shows little.

9 Conclusion and future work

We built an adaptive interviewer that estimates skill levels, stops on its own uncertainty, and explains wrong answers with evidence from the candidate's own words, while leaving the hiring

decision to a human. On 22 simulated candidates, its belief model made misconception findings about three times as trustworthy as a version without it, and adaptive question choice placed candidates better than a fixed list at equal length. The full system did not combine both advantages: probing beliefs cost level accuracy and length. The most useful next steps are to keep belief-probing questions out of the level estimate, to stop probing a belief once it is confirmed, to require a clear margin from the nearest cut point before stopping, to finish the run on all 60 candidates, and to complete the human checks.

Reproducing this report

Code, question bank and candidates are at `https://github.com/Srinjoy003/LMA-RAG`. The final run is `reviewed-main-v2`, produced by `scripts/run_rotating.py` in the order of `docs/final_run_order.txt`; its manifest records the code, bank, prompts and models. Every result number and figure in this report is generated by `report/make_final_figures.py` from the 22 complete candidates, and `scripts/evaluate.py` gives the full metric table. Every LLM response is cached on disk, so re-running from the cache gives the same interviews.

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